Tutorial Sheet 2

Introduction to Numerical Methods for ODEs and PDEs

Initial Value Problems, Numerical Stability & Accuracy

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Part I: Practical Exercises

Exercise 1 (Explicit Euler to solve the model problem)
Consider the model problem

$$y' = \lambda y\,, \quad y(0) = y_0\,.$$

We wish to obtain the solution $y(t)$ of this initial value problem from the initial time $t = 0$ to the final time $t = t_f$. Write a code to implement Explicit Euler, i.e.

$$y_{n+1} = y_n + hy'_n = y_n + h\lambda y_n\,,$$

where h is the step size.

1. *What is the exact solution $\tilde{y}(t)$ of the problem (in terms of λ)?*
2. *Select any initial condition y_0. Suppose $\lambda = -1$ and let $t_f = 25$. In the following we will compare the effect of different step sizes on the performance of explicit Euler for this problem. You may wish to plot the exact solution $\tilde{y}(t)$ along with the numerical solutions for comparison.*
   (a) *What is the maximum stepsize h_{max} possible to ensure numerical stability?*
   (b) *Using explicit Euler, compute and plot (as a function of time) the numerical solution with $h = h_{max}$.*
   (c) *Using explicit Euler, compute and plot (as a function of time) the numerical solution with $h = h_{max}/2$.*
   (d) *Using explicit Euler, compute and plot (as a function of time) the numerical solution with $h = h_{max} + 0.1$.*
   (e) *For each of the step sizes, plot the global error $\left|y_n - \tilde{y}(t_n)\right|$ as a function of the time step.*
   (f) *Comment on the numerical stability of explicit Euler for the different choices of h.*
   (g) *Compare the accuracy achieved for each selection of h. What is the cost of improved accuracy?*
3. *Suppose $\lambda = i\lambda_I$ is purely imaginary.*
   (a) *For some value of λ (you pick!), use explicit Euler to see (as we demonstrated in the lecture) that this particular numerical method is unstable for purely imaginary λ.*

Exercise 2 (Improved performance by including higher order derivatives)
Consider again the model problem

$$y' = \lambda y\,, \quad y(0) = y_0\,.$$

Instead of using explicit Euler, we wish to include the second derivative in the Taylor series expansion of y_{n+1} about $t = t_n$ in the hope of getting a more accurate numerical method.

1. *Derive the numerical method for solving the model problem, which includes the second derivative y'' in the Taylor series expansion of y_n about $t = t_n$.*
2. *Assess the stability properties of the numerical scheme.*
3. *What is the order of the local error of this scheme?*
4. *What is the order of the global error of this scheme?*
5. *Let $\lambda = -1$ and $t_f = 25$. Writing your own code or using the code provided, for some time step h, plot the numerical solution and compare it to the numerical solution achieved using explicit Euler in the previous exercise.*

Exercise 3 (Implicit Euler to solve the model problem)
Consider again the model problem

$$y' = \lambda y\,, \quad y(0) = y_0\,.$$

Write code to implement implicit Euler, i.e.

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}) = y_n + h\lambda y_{n+1}\,.$$

1. *Assess the numerical stability of this scheme for the model problem.*
2. *Select an initial condition y_0. For $\lambda = -2$, we wish to obtain a numerical solution using implicit Euler.*
   (a) *Let $t_f = 20$ and $h = 1.1$. Plot the numerical solution obtained using implicit Euler. How does the result compare to the numerical solution you would have obtained using explicit Euler with the same parameters?*
   (b) *Let $t_f = 20$ and $h = 0.1$. Plot the numerical solution obtained using implicit Euler. How does the result compare to the numerical solution obtained (using implicit Euler) in part (a)?*
3. *For $\lambda = i$, we wish to obtain a numerical solution using implicit Euler.*
   (a) *Let $t_f = 20$ and $h = 1$. Plot the numerical solution obtained using implicit Euler.*

(b) *Let* $t_f = 20$ *and* $h = 0.1$. *Plot the numerical solution obtained using implicit Euler.*

(c) *Comment on numerical stability and accuracy of the numerical schemes in parts (a) and (b).*

Exercise 4 (An example using Explicit Euler and Runge-Kutta)

We wish to obtain a numerical solution of

$\frac{dx}{dt} = -x + \sin(\omega t)$ $\therefore \frac{dx}{dt} + x = \sin(\omega t)$

$$\dot{x} = -x + \sin(\omega t)\,, \quad x(0) = 10\,,$$

Let $I(t) = e^{\int dt} = e^t$

$\therefore \frac{dIx}{dt} = e^t \sin(\omega t)$ $\therefore Ix = \frac{e^t}{1+\omega^2}[\sin(\omega t) - \omega\cos(\omega t)]$

$\therefore x = \frac{1}{1+\omega^2}[\sin(\omega t) - \omega\cos(\omega t)]$

where $\omega \in \mathbb{R}$ *is a scalar parameter.*

1. *Using either your own code or the code provided, use explicit Euler, with step size* $h = 2$, *to obtain a numerical solution of the ODE, with* $\omega = 2$. *Plot the resulting numerical solution and comment on the result.*

2. *Use explicit Euler, this time with step size* $h = 0.5$, *to obtain a numerical solution of the ODE, with* $\omega = 2$. *Plot the resulting numerical solution.*

3. *Using either your own code or the code provided, use the fourth-order Runge-Kutta, with step size* $h = 0.5$, *to solve the ODE, again with* $\omega = 2$. *Plot the resulting numerical solution and comment on the difference in accuracy between the numerical solution obtained using explicit Euler and fourth-order Runge Kutta.*

4. *Using explicit Euler again, this time with step size* $h = 0.1$, *obtain a numerical solution of the ODE. Plot the resulting numerical solution. Comment on the accuracy of the solution.*

5. *Suppose now the parameter* ω *is increased. How does this impact the selection of the step size* h *of a numerical solution? (You may wish to see this in practice, by experimenting with different values of* λ *and* h *in your code.)*

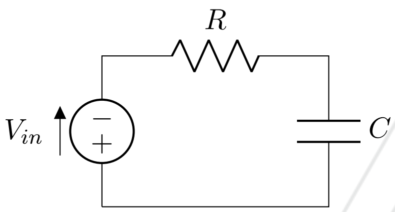

Figure 1: Schematic of RC circuit.

Exercise 5 (An example using the trapezoidal method)

The RC circuit shown in Figure 1 has input voltage $V_{in}(t)$. The voltage across the capacitor is the output voltage $V_{out}(t)$. For a given input voltage, the output voltage is the solution of the differential equation

$$\frac{dV_{out}}{dt} + \frac{V_{out}}{RC} = \frac{V_{in}}{RC} \,.$$

$V_{out}' = -\frac{V_{out}}{10} + \frac{\sin(t)}{10} + \frac{\sin(400t)}{10}$

Let $R = 100\ k\Omega$ and $C = 100\mu F$. Using either your own code or the code provided, use the trapezoidal method to obtain a numerical solution for the output $V_{out}(t)$ given the input

$$V_{in} = \sin(t) + \frac{1}{2}\sin(400t) \,.$$

Experiment with different step sizes.

Note that you can use an in-built solver, e.g. "fsolve", to obtain the solution of the algebraic equation required at each time step.

Exercise 6 (The simple harmonic oscillator - a second order ODE)
A simple harmonic oscillator is described by a second order ODE of the form

$$mx'' + kx = 0 \,,$$

where k and m are (scalar) constants. Defining the vector $y = \begin{bmatrix} x \\ x' \end{bmatrix}$, the second order ODE can be written as the system of first ODEs

$$y' = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} y = Ay \,.$$

Suppose the constant parameters are such that $k = m$.

1. *Compute the egeinvalues and the normalized eigenvectors of the matrix A.*
2. *Let Λ denote the diagonal matrix with the eigenvalues of A on its diagonal and let P denote the matrix with columns corresponding to the eigenvectors of A. Define the new vector*

$$z = \begin{bmatrix} x \\ x' \end{bmatrix} = P^{-1}y \,.$$

$Au = \lambda u = \lambda I u \quad \therefore (A - \lambda I)u = 0 \quad \therefore \det\begin{vmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\lambda \end{vmatrix} = \lambda^2 + \frac{k}{m} = 0 \quad \therefore \lambda = \sqrt{-\frac{k}{m}} = \pm i\sqrt{\frac{k}{m}}$

$\therefore \begin{bmatrix} \pm i\sqrt{\frac{k}{m}} & 1 \\ -\frac{k}{m} & \pm i\sqrt{\frac{k}{m}} \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0 \quad \therefore \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -i\sqrt{\frac{k}{m}} & i\sqrt{\frac{k}{m}} \end{bmatrix}$

for $k=m$.

$$\Lambda = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad P = U = \begin{bmatrix} \overset{\frac{\sqrt{2}}{2}}{\cancel{1}} & \overset{\frac{\sqrt{2}}{2}}{\cancel{1}} \\ \underset{-\frac{\sqrt{2}}{2}i}{\cancel{-i}} & \underset{\frac{\sqrt{2}}{2}i}{\cancel{i}} \end{bmatrix}$$

$$z = P^{-1}y \qquad \therefore z' = P^{-1}y' = P^{-1}Ay = P^{-1}P\Lambda P^{-1}y = \Lambda z$$

Show that

$$z' = \Lambda z \,. = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \qquad \therefore \begin{aligned} z_1' &= \lambda_1 z_1 \\ z_2' &= \lambda_2 z_2 \end{aligned}$$

3. *If the initial conditions are $x(0) = x_0$ and $x'(0) = x_d$, what are the corresponding initial conditions for $y(0)$ and $z(0)$?*

4. *The ODE defined in terms of z essentially consists of two decoupled first order linear differential equations. Select an appropriate scheme and obtain the numerical solutions which approximate $z_1(t)$ and $z_2(t)$ (where z_1 and z_2 are the two components of z.*

5. *Using the numerical solution obtained in part 4. of this exercise construct a numerical solution for $y(t)$.*

Part II: Theoretical Considerations

Exercise 7 (Trapezoidal Method - Linearization)
Consider the Trapezoidal method for obtaining a numerical solution of the initial value problem

$$y' = f(t, y)\,, \quad y(0) = y_0\,.$$

In general, this multi-step method requires solving a nonlinear algebraic equation at each time step.

1. *Compute the linearization of $f(t_{n+1}, y_{n+1})$ about y_n.*

2. *Show that applying the trapezoidal method to the linearized problem (*i.e. *when $f(t_{n+1}, y_{n+1})$ is replaced with the linearization computed in part 1. of this exercise) results in a numerical solution with the same order of accuracy achieved when applying the trapezoidal method to the original (nonlinear) problem.*